A PRELIMINARY STUDY ON THE DETECTION OF GLACIAL LAKE OUTBURST FLOOD IN NORWAY USING SENTINEL 2 IMAGERY

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***Abstract*—Over recent decades, glacier retreat has altered dynamics and increased hazards, such as glacial lake outburst floods (GLOFs), leading to severe downstream flooding. While GLOFs are typically detected through on-site monitoring, existing remote sensing methods primarily map glacial lakes, without identifying GLOF events. This paper proposes an unsupervised approach to detect GLOFs using Sentinel-2 image time series. We train a deep network to model non-draining lake time series, preventing them from appearing as anomalies when drained. The model, based on convolutional Long-Short-Term Memory and a 3D convolutional neural network, reconstructs non-draining lake time series. Inference errors reveal draining lakes. Preliminary experiments on Norwegian glacial lakes show promising results for automated GLOF detection.**

***Keywords*—Anomaly detection, recurrent neural network, convolutional LSTM, 3D-convolutional neural network, GLOF event detection.**

I. Introduction

Over the past few decades, the rapid retreat of glaciers has significantly altered glacier dynamics, contributing to an increase in hazards, including glacial lake outburst floods (GLOFs) [1], [2]. GLOFs occur when glacier meltwater is suddenly and catastrophically released, causing vast flooding downstream [3]. GLOF events are mainly detected via in-situ glacial lake monitoring (which is extremely expensive) [4], and physical science [5], [6]. Automatic methods in the remote sensing literature focus on mapping glacial lakes with a surface larger than 0.01 km^2 [7], which are the minority of those prone to GLOF events, especially in Norway. Recent works generate a more comprehensive glacier and glacial lake inventories of Norway using a standardized semi-automated band ratio followed by manual refinement to derive glacier outlines and ice-marginal lakes (i.e., bodies of freshwater that form along the edges of glaciers) [8], [9], without focusing on GLOF events. Hence, to enrich these inventories, automatic detection of GLOF events in glacial lakes using remote sensing data is needed. The method should consider the limited number of GLOF events and the acquisition scenarios (e.g., lakes in deep valleys and high latitudes).

Given the limited number of recorded GLOF events, their automatic detection requires methodologies that do not need a large and balanced training dataset, such as anomaly detection (AD). AD models the standard behavior of the dataset and identifies anomalies in those cases that significantly deviate from it. AD is used in Remote Sensing (RS) image time series (TSs) to detect floods [10], agronomic anomalies [11], and dam collapse [12]. However, no work has been done to automatically detect GLOF events. Moreover, most literature methods for time series anomaly detection employ hand-crafted features, which can lead to suboptimal performance. Deep Learning (DL) AD methods for time series typically leverage the pattern identification capabilities of convolutional layers and the temporal modeling capabilities of Long-Short Term Memory (LSTM). The latter models the temporal behavior of standard examples while emphasizing the anomalies in the TSs [13], [14]. However, most AD methods focus on temporal information without exploiting spatial information. Spatial information improves performance when anomalies occur in the spectral signature variation over time and when objects change in space, such as in our case.

This paper proposes an unsupervised anomaly detection method that analyzes TSs of Sentinel 2 imagery to identify GLOF events. We utilize a deep network to model the spatiotemporal behavior of non-draining glacial lakes, serving as the standard behavior that the network should learn. Conversely, the TSs with GLOF events representing draining lakes have unexpected temporal behavior that the proposed method

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should identify as anomalies. We propose to unsupervisedly train a temporally and spatially aware deep network consisting of Convolutional LSTM (ConvLSTM) and 3D Convolutional Neural Network (3D-CNN) with non-draining TSs. The ConvLSTM models the temporal behavior of the glacial lakes during the year, while the 3D-CNN extracts spatially meaningful information. In the inference, the reconstruction of draining glacial lakes TSs with unexpected temporal behavior provides a larger reconstruction error than that of non-draining lakes. The reconstruction error is analyzed to distinguish between non-draining and draining glacial lakes, thereby identifying GLOF events. We successfully applied the proposed method to Sentinel-2 imagery to identify GLOF events in Norwegian glacial lakes of varying dimensions.

II. Proposed method for GLOF event detection

This paper proposes an anomaly detection method that analyzes Sentinel-2 image TSs T composed of N monthly images $H \times W \times S$, where H is the height, W is the width, and S is the number of spectral channels. We propose using a deep architecture to model the spatiotemporal behavior of glacial lakes and distinguish between non-draining (ω_{nd}) and draining (ω_d) glacial lakes. Given the limited number of TSs with GLOF events, the proposed method addresses GLOF event detection as an unsupervised anomaly detection problem. TSs without the GLOF events are the standard behavior the network should learn. Conversely, the TSs associated with GLOF events exhibit unexpected behavior compared to non-draining lakes, which identifies them as anomalies.

Fig 1 shows the flow of the proposed method. Let us assume having a dataset X composed of I yearly TSs $T_i \in X$, $i = 1, \ldots, I$ with the Sentinel-2 band 11 and Normalized Difference Water Index (NDWI) obtained from the Sentinel-2 spectral bands. We used band 11 and NDWI, as they have been proven to characterize best GLOF events (see Section II-A). We assume that X is mainly composed of non-draining examples. We train a temporally and spatially aware deep network based on 3D-CNN and ConvLSTM using a X. The network is trained to simultaneously process the temporal and spatial domains, extracting meaningful features that model the yearly behavior of non-draining lakes. In the inference, the reconstruction of the draining TSs is expected to show a more significant reconstruction error than that of the TSs of non-draining lakes. The reconstruction error is analyzed to discriminate ω_{nd} and ω_d and identify GLOF events.

A. GLOF event spectral analysis

From a spectral perspective, GLOF events are marked by an increase in the reflectance of bands 11 (see Fig.2). Prior to the GLOF event, the spectral signature of this band decreased during spring and early summer because of the ice melt and the presence of water. Following a GLOF event, the reflectance stabilizes and then gradually increases with the onset of winter as the water begins to freeze. In contrast, NDWI increases in late spring and early summer as the ice melts, then decreases during the GLOF event (see Fig.2). This spectral behavior is typical for many glacial lakes affected by GLOFs. However, several challenges arise in more complex scenarios. Lakes at higher latitudes may remain covered by snow or ice; terrain and cloud shadows can be misclassified as water during detection, and persistent cloud cover can obscure glacial lakes altogether [7], [15]. Additionally, lakes that do not fully drain may exhibit only subtle changes in spectral response. Therefore, the proposed method must be robust enough to capture GLOF dynamics under non-ideal conditions.

B. Network architecture and training

To simultaneously learn the temporal and spatial behavior of glacial lakes, we propose a UNet-based architecture consisting of an encoder and a decoder. The encoder learns spatial context features from the input TSs and compresses the spatial information to increase the receptive field. The encoder output is processed by a ConvLSTM that simultaneously learns the spatial context information and temporal behavior of the TSs. The decoder input is the ConvLSTM output and learns more complex spatiotemporal features and derives the reconstructed input $\hat{X}$. The encoder and decoder comprise 3D convolutional layers followed by batch normalization and ReLU activation functions. The 3D convolutional layers are organized into groups of three layers, following the bottleneck strategy to minimize the number of parameters. The bottleneck layer kernel size is $k \times k \times 1$, focusing the network on spatial information but not temporal information. The model is unsupervised trained for e epochs with a Root Mean Squared Error (RMSE) as loss function:

$$RMSE(\hat{X}, X) = \sqrt{||\hat{X} - X||_2^2}. \tag{1}$$

C. GLOF event detection

To detect GLOF events, each TS T is reconstructed $\hat{T}$ with the deep network. We compute the reconstruction error $RMSE(\hat{T}, T)$, and we detect anomalous T (i.e., a GLOF event) by applying a threshold th. Similarly to [13], th is computed as follow:

$$th = \frac{1}{I}\sum_{i}^{I} RMSE(\hat{T}_i, T_i) + \sqrt{\frac{1}{I}\sum_{i}^{I}(RMSE(\hat{T}_i, T_i) - \mu)^2} \tag{2}$$

where μ is the mean value of RMSE in (1). The GLOF event is identified when $RMSE(\hat{T}, T)$ is greater than th. Conversely, T represents a non-draining lake.

III. Dataset and Experimental Results

To validate the proposed method, we generated a strongly unbalanced dataset of TSs, consisting mostly of non-draining and a few draining glacial lakes in Norway, acquired between 2016 and 2023. We first identified 274 Norwegian glacial lakes using existing inventories [16] and a lake detection algorithm based on the Normalized Difference Water Index (NDWI). Since 2016, a total of 24 GLOF events have been recorded, originating from six different lakes [6]. Secondly, we selected lakes larger than $0.003\,\mathrm{km}^2$ (more than 300 pixels in the

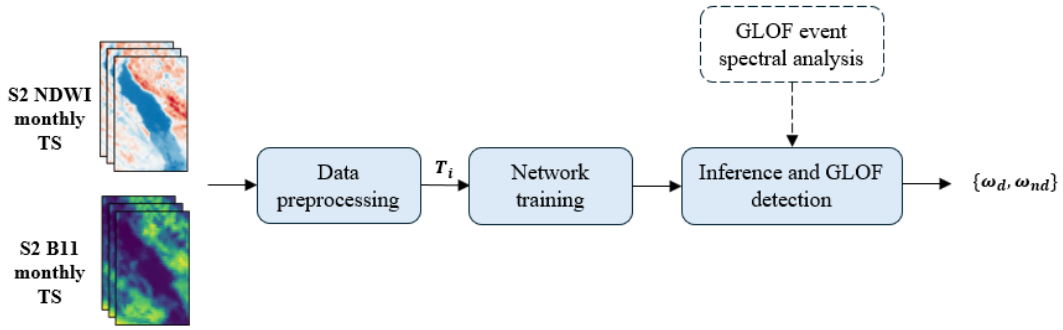


Fig. 1: Flow chart of the proposed method for GLOF event detection.

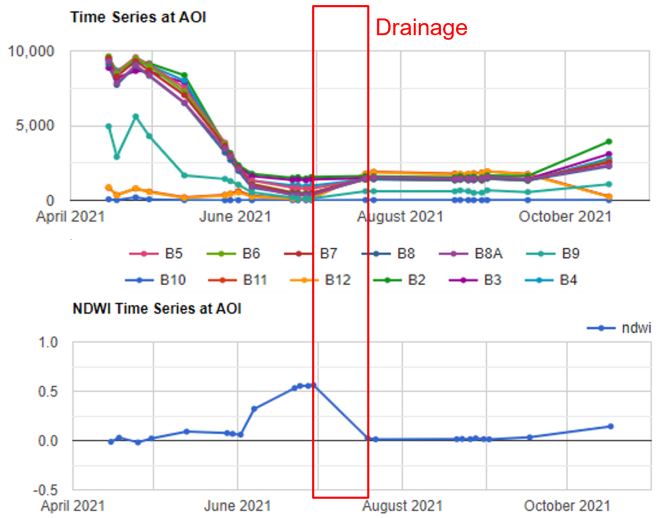


Fig. 2: Spectral analysis of the summer TS of a glacial lake with a GLOF event highlighted in red.

Sentinel 2 image). This reduces the dataset to 12 TSs of draining and 268 TSs of not-draining glacial lakes. Then, for each glacial lake, we generated the annual TSs with monthly median composites of the NDWI and Band 11. Data gaps were interpolated with images from the previous or following month. Each TS was masked to highlight the lake and then normalized to the range of [0, 1] to preserve the temporal TS behavior. We sampled these lakes to obtain 1331 image TSs of non-draining lakes and 12 TSs of draining lakes (GLOF events). Each TS comprises ten images with a width and height of 256×256 and two channels, NDWI and Band 11. The non-draining samples were divided into training and test sets, each comprising 1053 and 278 samples, respectively. The proposed method was validated using the non-draining test set and the 12 TSs of draining lake samples.

The DL model was trained for $e = 200$ epochs with a learning rate of $1e^{-4}$ using the Adam optimizer with a batch size of 16. We used a kernel size $k = 3$ to minimize the number of model parameters. We used the Overall Error (OE), recall/accuracy, False Alarm (FAs), and Missed Alarm (MAs) rates to quantitatively evaluate AD performance.

TABLE I. QUANTITATIVE RESULTS OF THE PROPOSED METHOD.

	FAs (%)	MAs (%)	OE(%)	Recall (%)
Proposed method	8.82	16.66	9.16	83.34

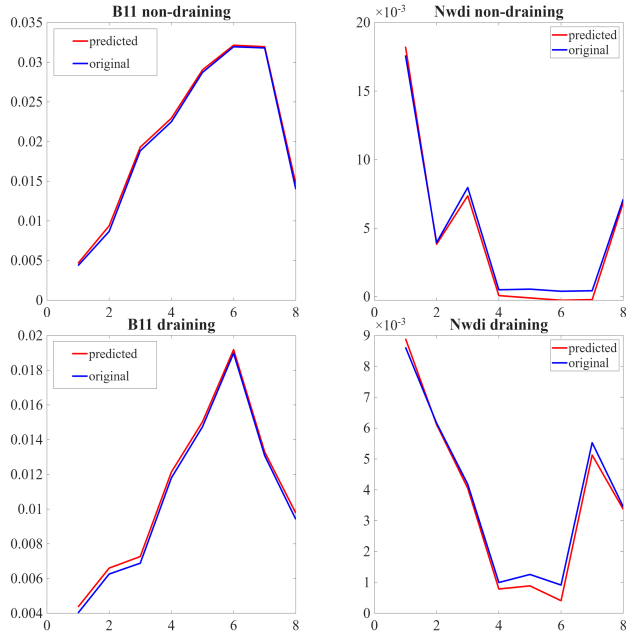


Fig. 3: Example of time series trend for the average value of band 11 and the NWDI index for a non-draining lake (top) and a draining lake (bottom).

Table I shows that the proposed method achieved promising results, detecting most GLOF events with a recall equal to 83.34%. However, the resulting FA of 8.82% and the OE of 9.16% highlight the network challenges in generalizing the temporal behavior of the data non-idealities (e.g., acquisition conditions and different geographical locations).

Fig 3 shows that the proposed DL model can accurately reconstruct the temporal behavior of a non-draining lake NDWI with a small error. As expected, the reconstruction of the NDWI temporal behavior of a draining lake has a larger error than that of a non-draining lake. This is also

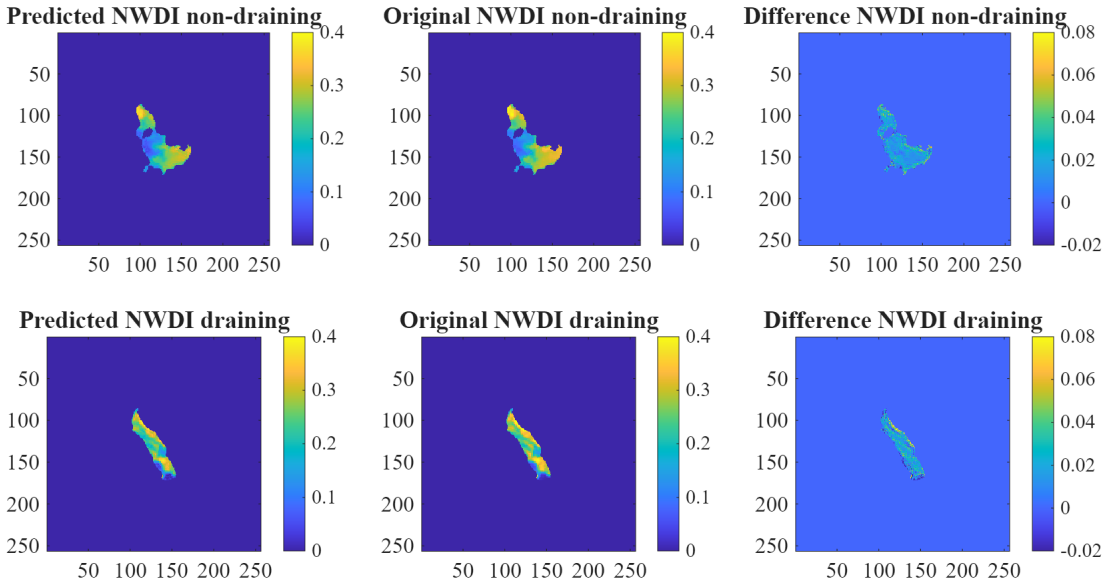


Fig. 4: Examples of the original and predicted image for the 5th image of the series (that after the GLOF event) of a non-draining lake (top) and a draining lake (bottom).

visible in Fig 4, where the NDWI of the non-draining lakes is reconstructed with an extremely low error (e.g., 0.006). Conversely, the NDWI of a draining lake is reconstructed with a large error (e.g., 0.015) as the network expects water in the lake, as in a non-draining case seen during training.

IV. Conclusions and Future Works

We proposed an unsupervised anomaly detection method for identifying GLOF events using a DL model based on 3D-CNN and ConvLSTM. The proposed model simultaneously learns the temporal and spatial behavior of non-draining lakes in Sentinel-2 TSs. It utilizes the RMSE to identify unexpected spatiotemporal behavior in the TS reconstruction associated with GLOF events. The preliminary results show that the proposed method accurately reconstructs the non-draining glacial lake TS with an extremely low error compared to the draining TSs. The proposed method successfully detected most of the GLOF events, despite a strongly unbalanced and challenging dataset. However, the method has many FAs due to the high variability of non-draining lake temporal behavior.

In future work, we plan to reduce the false alarm rate by extending the method to the multi-modal processing of Sentinel-1 and Sentinel-2 image TSs, while improving the GLOF event detection in different acquisition scenarios (e.g., cloudy environment). Sentinel-1 data enables more accurate discrimination between water and ice, increasing the temporal resolution of the image time series and thereby improving the temporal behavior modeling. We also plan to validate the method on glacial lakes in Alaska, which exhibit varying sizes and geographical properties.

References

[1] G. Zhang *et al.*, "Characteristics and changes of glacial lakes and outburst floods," *Nature Reviews Earth & Environment*, vol. 5, no. 6, pp. 447–462, 2024.

[2] R. Hock and M. Truffer, "Glacier hazards: Will they change in the future?" *Arctic, Antarctic, and Alpine Research*, vol. 56, no. 1, p. 2335991, 2024.

[3] L. Iturrizaga, "Glacier lake outburst floods," *Encyclopedia of snow, ice and glaciers*, pp. 381–399, 2011.

[4] J. Yang *et al.*, "The role of satellite remote sensing in climate change studies," *Nature climate change*, vol. 3, no. 10, pp. 875–883, 2013.

[5] A. Emmer *et al.*, "Progress and challenges in glacial lake outburst flood research (2017–2021): a research community perspective," *Natural Hazards and Earth System Sciences Discussions*, vol. 2022, pp. 1–34, 2022.

[6] N. Lützow *et al.*, "A global database of historic glacier lake outburst floods," *Earth System Science Data Discussions*, vol. 2023, pp. 1–27, 2023.

[7] S. Wangchuk and T. Bolch, "Mapping of glacial lakes using sentinel-1 and sentinel-2 data and a random forest classifier: Strengths and challenges," *Science of Remote Sensing*, vol. 2, p. 100008, 2020.

[8] L. M. Andreassen *et al.*, "Inventory of norwegian glaciers," *Rapport*, vol. 38, 2012.

[9] ——, "An inventory of norway's glaciers and ice-marginal lakes from 2018-19 sentinel-2 data." *Journal of Glaciology*, vol. 68, no. 272, 2022.

[10] Z.-G. Zhou *et al.*, "Detecting anomaly regions in satellite image time series based on seasonal autocorrelation analysis," *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 3, pp. 303–310, 2016.

[11] K. M. Leon-Lopez *et al.*, "Anomaly detection and classification in multi-spectral time series based on hidden markov models," *IEEE transactions on geoscience and remote sensing*, vol. 60, pp. 1–11, 2021.

[12] V. L. Gino *et al.*, "Integrating unsupervised machine intelligence and anomaly detection for spatio-temporal dynamic mapping using remote sensing image series," *Sustainability*, vol. 15, no. 6, p. 4725, 2023.

[13] Y. Zhang *et al.*, "Unsupervised deep anomaly detection for multi-sensor time-series signals," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 2, pp. 2118–2132, 2021.

[14] A. Kulshrestha *et al.*, "Use of lstm for sinkhole-related anomaly detection and classification of insar deformation time series," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 15, pp. 4559–4570, 2022.

[15] J. M. Abderhalden *et al.*, "Tracking rapid and slow ice-dammed lake changes through optical satellites and local knowledge: a case study of tystigbreen in norway," *Journal of Glaciology*, vol. 70, p. e71, 2024.

[16] "Glacier lake outburst flood (glof)," accessed: 2025-07-10. [Online]. Available: https://glacier.nve.no/Glacier/viewer/GLOF/en/